Smart Censorship: Real-Time Obscene Content Detection Using Skin Pixel Aggregation and Humanoid Shape Recognition

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***Abstract*—Over-the-top (OTT) platforms have captured a large share of the market and offer streaming media services directly to viewers using the internet. Ever since the world has felt the pain of the COVID-related lockdown, these platforms have garnered market share at an exponential rate. A few years after these platforms were introduced, they successfully dethroned television through cable and satellite. This has come with vulnerabilities, as all Internet content has come to the television, and this requires careful content blocking in a country like India, where we, the people, still adore our family and value our culture. An attempt has been made in this proposed work to detect and flag offensive and obscene content. The proposed obscene frame detection approach model can produce satisfactory results. The average of eight different videos of short durations demonstrates a recall of 96.04% and a below 3% false positive rate on the test set. The algorithm produces results with 96.23% precision and 98.55% accuracy.**

Keywords: Obscene Content Detection, Skin Pixel Aggregation, Humanoid Shape Recognition, OTT Platform Regulation, Real-Time Video Analysis

I. INTRODUCTION

The COVID-19 pandemic and its resulting lockdowns led to an unprecedented surge in digital content consumption across mobile and television platforms. With education shifting online, children in India gained increased and often unsupervised access to mobile devices. While essential for remote learning and entertainment, this shift introduced significant vulnerabilities, particularly the unfiltered exposure to all types of internet content, including explicit or culturally inappropriate material.

In India, a society deeply rooted in family values and cultural norms, the influx of unregulated streaming content through OTT platforms such as Netflix, Amazon Prime Video, and Disney+ Hotstar raises critical concerns. Today's OTT platforms initially operated without the oversight of any regulatory authority compared to traditional cinema and cable television. To date, the Indian government shows no signs of accepting responsibility for controlling such services. Ever since, there have been consultations by the Ministry of Information and Broadcasting with relevant parties to draw up policies and set up a control system. These initiatives focus on the prevention of content that is sexually obscene, pornographic, illegal, or otherwise harmful to children [2]. Title-based classification methods, which have been previously employed to filter content, lack effectiveness in India. There are numerous ways malicious content creators can disguise their true intentions and slip unnoticed. Even language-based filters, such as the one proposed by Singh and Kumar [1] using Devanagari script for title recognition, do not escape this misleading trend. Regardless of how advanced the regulations are, there remains a critical technical problem: automatic and precise identification of obscene visual content in video streams or close to real-time is still a challenge.

OTT platforms require robust, scalable, and intelligent mechanisms that can scan, analyse, and flag inappropriate frames without relying solely on text-based metadata or manual screening. Existing solutions fail to address content moderation visually, especially in culturally diverse environments like India. Furthermore, many currently available detection systems either exhibit high false positive rates or struggle to handle the diversity of skin tones, lighting conditions, and ambiguous scenes in real-world video content. This paper presents a real-time obscenity detection framework tailored for OTT video content to address these gaps. The core contributions of this work include:

- A robust frame-based detection model that utilises Histogram of Oriented Gradients (HOG) and Support Vector Machine (SVM) for humanoid shape detection.

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- An advanced skin pixel analysis approach that combines multiple colour spaces (RGB, Normalised RGB, HSV) to handle diverse skin tones and improve detection accuracy.
- A rule-based obscenity classifier that flags frames as obscene based on thresholds derived from skin pixel ratios, intensity values, and region aggregation logic.
- A test of practical video clips, with an average score of 98.55 per cent, precision of 96.23 per cent, and a very low false positive rate registering less than 3 per cent, confirming the implausibility of the framework to be employed in real-time.

The study, therefore, becomes part of the continuous process of safer, culture-sensitive intuition in digital content consumption and acts as a base model in future incorporations in OTT platforms, combined with the process of regulation.

II. Related Work

A. Language and Textual Content Recognition in Indian Context

As in the case of many violations of community standards, the analysis starts with a video's title and other associated text. In terms of the Indian problem, the multilingual nature of the country is particularly interesting. Singh and Kumar [2] applied connected component labelling segmentation with Histogram of Oriented Gradients (HOG) based character recognition to a fragmentary model of Devanagari script. Singh [3] and Kumar also designed a classification model for texts of Indian languages, achieving 93.3\% mean accuracy in the benchmarks. With [16], Krishnarao et al. extended this model using Conditional Random Fields (CRF) and Support Vector Machines (SVM) to perform named entity recognition for Hindi. They claimed that feature language independence would allow these to be used in other languages and scripts like Telugu, Bengali, and Marathi. These methods, albeit useful, would be outright useless in combating deliberate foul titling by unscrupulous uploaders using text disguise, since metadata filters fail to detect obscenity. Text-based filters are less effective in systems like OTT media due to thin layers of misleading metadata disguised as titles.

B. Skin and Nudity Detection Techniques

Detecting nudity in images has been thoroughly researched, with an approach that starts with skin region identification. Rigan [4] noted the shortcomings of single-colour spaces' inaccuracy. To perform better, he proposed a skin detection model based on RGB, Normalised RGB, and HSV colour spaces using linear regression. This multi-space approach sought to improve detection robustness by accommodating diverse human skin tones. Santos et al. [5] introduced zoning strategies that involved partitioning an image into regions and extracting texture and colour features from each. They reported better classification performance when combining two colour-based and two texture-based descriptors with SVM for classification. Emon and Rahat [14] proposed a real-time skin detection system built upon the percentage of skin pixels and facial areas. Coupled with face recognition using Google Vision API, they were able to achieve 95% accuracy with a low false positive rate using HSV and RGB colour spaces. Platzer et al. [13] designed a model that locates contiguous skin regions in images and subsequently classifies them with SVM. Their work underscored the importance of geometrical characteristics and grouping in accurate nudity classification. Nevertheless, their technique witnessed a low recall score of 65.7, which means that most nude images were not flagged.

C. Obscene Content Detection in Images and Videos

The problem of identifying explicit content in full video streams becomes more difficult than in still images because of movements, changing light, and the complexity of scenes. Ion and Cristian [7] proposed a nudity classification model using body part segmentation (e.g., chest, thighs, back) and assigning nudity scores via multi-label classification. Ahuja et al. [10] further emphasised the importance of controlling the spread of Not Safe for Work (NSFW) content by identifying nude zones such as genitals, breasts, and buttocks from annotated image datasets. Using public datasets, Gangwar et al. [8] performed skin detection alongside deep learning-based pornography detection method empirical comparisons. They stated that traditional methods do not perform well with differing skin tones and background clutter. AlDahoul et al. [9] used a deep learning-based CNN for feature extraction and an SVM for the classification problem. They also worked on object detection with YOLO, and a ResNet101–Random Forest hybrid was reported to achieve 90.03% F1-score and 87.75% accuracy. Using artificial neural networks, Zhelonkin and Nikolay [11] developed a solution for nudity in non-human images, like cartoons and animated characters. Their work addressed the concern of child-facing media platforms by extending nudity detection beyond real-life scenarios. During the SemEval-2022 challenge, Gu et al. [15] investigated detecting misogyny in memes. They described the use of a RoBERTa-based classifier for text obscenity detection, demonstrating the need for moderation on multiple content levels—textual and visual. Gandhi et al. [12] studied obscenity and offensiveness in images and advertisements related to e-commerce. Object detection and image classification were used to mitigate biased consumer behaviour and legal concerns. Their study brought to light the danger presented by visual content in web business environments.

D. Challenges in Current Detection Methods

Despite innovations in processing images and videos, several weaknesses remain in current approaches. Many traditional skin detection models struggle to generalise across diverse skin tones and backgrounds, often producing high false positive or false negative rates. Deep learning-based models require large, annotated datasets and are

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computationally intensive, limiting their real-time applicability. Another gap lies in the lack of a post-detection strategy. As Tariq et al. [6] noted, most works focus solely on detection without considering how platforms should act on the flagged content. They advocated for obscene content detection at the hardware level to stop such material from being created or shared in the first place. Additionally, most literature fails to account for cultural nuances, especially in countries like India, where cultural sensitivities vary widely. There is a pressing need for models that detect nudity and assess context, differentiating between artistic, educational, or explicitly offensive content.

In a nutshell, the proposed literature review can be significant when it comes to language-based filtering, skin detection, and nudity classification. However, putting these methods together into an up-scaled and culturally enterprising system is a thornier issue. The presented work tries to close the specified gap by incorporating humanoid form detection into the multi-colour space skin pixels analysis and contextual thresholding of the obscene content in the video. In the section that follows, the proposed methodology is expounded on.

III. Proposed Methodology

A. Overview of the System Architecture

The proposed research is an attempt to allocate a hybrid strategy to find obscene material in video streams by using humanoid shape detection and skin pixel grouping. It is aimed at marking videos that might show nudity or indecent shots, depending on skin pixel percentages and distribution in the humanoid figures. Its pipeline involves frame extraction, object detection, skin pixel classification, region aggregation, and classification based on the threshold.

B. Frame Extraction and Preprocessing

The first step lies in decomposing the video into separate frames via applying the classic frame extraction algorithms. The individual frame is recorded and processed separately. This step ensures adequate temporal granularity when studying dynamic content as well as adequate resolution, which will be used in visual processing.

C. Humanoid Shape Detection Using HOG + SVM

To identify areas of interest, histograms of oriented gradients (HOG) descriptors are computed on every frame to extract features that can be likened to humans. The model detects and localises humanoid objects using a pre-trained SVM (Support Vector Machine) classifier. This precaution guarantees the possibility of using skin detection fewer times later, considering the results, in only areas that most probably depict individuals, reducing the possibility of a false positive, confounded with meaningless image portions.

D. Skin Pixel Detection

The proposed system uses a multi-model colour detection approach to identify and categorise skin pixels:

- **RGB, Normalised RGB, HSV Colour Models**: The pixel in the humanoid area is analysed in the form of RGB, normalised RGB and HSV colour models. The resulting combination of these models increases the resistance to the change in skin colour and the change in lighting conditions. A binary vote of whether a particular pixel is defined as skin or non-skin is made by comparing the colour qualities of each pixel to empirically established skin quality values.
- **YCbCr and HSV Merge:** For additional enhancement of the classification accuracy, the skin detection module combines the smart colour models, YCbCr and HSV. The YCbCr model decomposes luminance and chrominance differentials, which can be used to distinguish the skin region more easily, regardless of the complex light conditions. The HSV model helps in managing the variations of hues, which is convenient when it comes to discriminating against similar-colored clothes and real skin.

E. Region Aggregation and Bounding Polygon Computation

After classification, all adjacent skin pixels are grouped into contiguous regions. The three biggest of such areas are then chosen as ones to be studied further. Bounding polygons are drawn around these regions, and their areas are computed. The ratio of skin pixels within these polygons is calculated in relation to the total polygon area and the overall frame.

This regional analysis ensures that isolated patches of skin (e.g., visible arms or face) are not misclassified as nudity. Instead, obscene content is inferred only if the skin exposure surpasses specific thresholds in terms of size and concentration.

F. Threshold Conditions for Obscenity Flagging

The system flags a frame as obscene based on the following threshold conditions:

- Total Skin Area Threshold: The frame is considered potentially obscene when the adjacent skin region occupies more than 15% of the image area.
- Bounding Polygon Thresholds: The frame is not flagged when the largest skin region contributes less than 35% of the total skin pixels. The second and third largest regions must each contribute less than 30% to meet the non-obscene criterion.

A frame is classified as non-obscene when the number of skin regions exceeds 60% and the average intensity within the bounding polygons remains below 0.25. These criteria are designed to balance detection sensitivity with specificity, avoiding misclassification in partial or culturally acceptable exposure cases.

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IV. Dataset and Experimental Setup

A. Dataset Description

NRIA Person Dataset for Human Detection: The INRIA Person Dataset is employed to facilitate accurate humanoid shape detection using Histogram of Oriented Gradients (HOG). The data on this set is made up of grayscale and colour images with a standardised size of 64 128 pixels, which also fits the specification of an aspect ratio of 1:2 that is required to generate effective HOG descriptors. It is well known in terms of pedestrian detection applications and provides a great range of human pose variations, making the HOG-SVM classification pipeline more robust. Skin Dataset to Skin Pixel Classification: A selected skin dataset was to be incorporated so that meaningful skin pixel detection can be performed under different ethnic backgrounds and light orientations. This dataset includes annotated skin pixel data across a wide range of skin tones. It supports reliable region-based skin segmentation while minimising false positives arising from non-skin regions with similar colour profiles.

B. System Configuration

All experiments were conducted in a controlled laboratory environment on a system configured with an Intel i7 processor, 16 GB of RAM, and an NVIDIA GTX GPU to support accelerated processing. This configuration was selected to ensure the smooth execution of computationally intensive tasks such as frame extraction, human detection, and skin pixel analysis required by the proposed methodology.

C. Module Integration and Workflow

The system follows a modular approach, with individual components integrated sequentially to form the complete classification pipeline. The workflow is as follows:

- **Frame Extraction:** The input video is segmented into individual frames.
- **Humanoid Detection:** HOG descriptors are computed for each frame, and an SVM is used to detect humanoid regions.
- **Skin Pixel Classification:** Detected humanoid regions undergo skin detection using a trained skin classifier.
- **Region Aggregation and Polygon Computation:** Skin regions are grouped, and bounding polygons are generated.
- **Threshold application**: It plays a critical role in accurately classifying frames as obscene or non-obscene. In the proposed methodology, a frame is flagged as potentially obscene if the cumulative skin coverage exceeds 15% of the total frame area. However, to avoid false positives, additional conditions are applied. If the largest skin region accounts for less than 35% of the total detected skin pixels, and the second and third largest regions each contribute less than 30%, the frame is classified as non-obscene. Furthermore, if the total number of skin regions exceeds 60% and the average pixel intensity within the bounding polygons is less than 0.25, the frame is also deemed non-obscene. These multi-layered thresholds ensure a balanced trade-off between sensitivity and specificity, enhancing the robustness of the detection system.

V. Results and Analysis

A. Execution Time and Real-Time Capability

The proposed system demonstrates promising real-time performance based on benchmarked execution times. The frame extraction process for a 2-minute and 6-second video was completed in under 520 milliseconds. The humanoid shape detection using the HOG descriptor and SVM classifier took less than 75 milliseconds, while the model loading time remained under 120 milliseconds. The skin detection and classification stage was completed in under 195 milliseconds. These timings attest to the fact that the suggested method is efficient and best fits in real-life or online detection of obscene content.

B. Video Clip-Wise Performance Analysis

To assess the performance and stability of the offered methodology, it is subjected to eight video clips with different scenes and varying degrees of skin exposure. The system's classification outcomes were evaluated against every clip based on commonly accepted evaluation measures: precision, recall (sensitivity), specificity, F1-score, and accuracy. These measurements were based on the components of the confusion matrix, mainly current positives (TP) and false negatives (FN). Table 1 contains the results of each separate video clip, with a more detailed insight into the system behaviour when processing various content types. This analysis, per clip, is imperative in determining how the system can tackle tough situations, together with consistency.

TP	FP	Prec (%)	Recl (%)	Spec (%)	F1 (%)	Acc (%)
120	4	96.77	99.1	98.63	97.91	98.77
118	0	99.9	98.33	99.00	99.16	98.83
118	4	96.72	98.33	98.63	97.52	99.1
106	8	92.98	88.33	98.26	99.6	98.36
120	2	98.36	100	98.81	99.17	98.83
114	4	96.61	95	98.53	95.8	98.17
116	2	98.31	96.67	99.11	97.48	98.5
110	12	90.16	91.67	98.19	90.91	97.17

The mean result of combining all eight short clips of videos also identifies that the suggested system is effective and reliable when identifying obscene content. The system recorded an accuracy of 96.23 percent, a recall (sensitivity) of 96.04 percent, and a total accuracy of 98.55 percent, and the false positive rate was less than 3 percent. These

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measures show that the system has great power to detect obscene frames correctly while reducing false alarms. The precision is high, which means that most of the frames flagged were obscene, and the recall is also high, which means that most of the obscene contents were found correctly. Despite the variation of performance within clips (a bit less precision in Clip 8), this variation points to real-world conditions, such as scenes with a large exposure of skin, but which are not necessarily obscene. Nevertheless, the model demonstrated good results on all video clips, proving its high stability and applicability to different situations at a video level.

C. *False Positives and Mitigation Strategies*

Overall, the system works well, but it yielded false positives in scenes that featured short-sleeved clothing and swimwear or scenes on the beach where skin is shown in high amounts but with no nudity. The model applies multi-layered thresholds on skin region distribution and average intensity to curb this. Such mechanisms cut down false alarms due to dispersed or naturally exposed skin. However, contextual misclassification is still a complication. In an improved version in the future, semantic understanding of the scene, pose, as well as attention-based deep learning, can be added to achieve a better understanding of intent and context to increase precision further, yet retain the same high sensitivity in real-world visual scenarios.

VI. Conclusion and Future Work

Over-the-Top (OTT) platforms have rapidly gained acceptance and transformed user access to digital content and streaming media services. Furthermore, this unrestricted access opens the gateway for the rampant dissemination of obscene material, especially for cultures like India, which are more sensitive to such content. Addressing such issues, this paper proposes a technique for the detection of obscene content based on video frame analysis of skin exposure and humanoid shape detection. The proposed model achieved promising results: an average precision of 96.23%, a recall of 96.04%, and an accuracy of 98.55% across 8 short video clips. A false positive rate of less than 3% attests to the system's reliability. The method performs well under varied conditions by leveraging HOG descriptors, SVM classification, and multi-space colour models (RGB, normalised RGB, HSV, YCbCr). Although real-time capability has been demonstrated on standard hardware, future improvements include integrating deep learning-based feature fusion, such as CNNs and transformer models. Incorporating this system with live-streaming platforms and edge devices may enable automated moderation. This work provides a strong foundation for AI-based censorship in digital media.

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